Surveillance Radar

Representative baseline datasheet for MATLAB radar modeling exercises

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Classification: Training Use Only

IMPORTANT - REPRESENTATIVE BASELINE ONLY.

This document reformats representative baseline values for classroom modeling. Search volume, scan rate, elevation limits, detection threshold, reporting range, and target model remain mission- and site-configurable.

S-band rotating AESA surveillance

Fixed-site, long-range primary radar combining mechanically rotated all-azimuth coverage with electronically controlled elevation scanning.

2.9	750	37
GHZ NOMINAL FREQUENCY	KW PEAK POWER	DBI PEAK GAIN

Configuration at a glance

Radar role	**Primary surveillance**
Installation	**Fixed site**
Architecture	**Rotating AESA**
Coverage	**Continuous 360 deg**
Waveform	**Coherent pulsed**
Configuration	**Baseline hardware**

1. System identity and operating envelope

Parameter	Representative value
System designation	Surveillance Radar (training representative)
Radar type	Fixed-site, monostatic, hybrid mechanically rotating AESA
Frequency band	S-band
Nominal frequency	2.9 GHz
Azimuth coverage	360 deg continuous
Azimuth scan method	Mechanical rotation; rate configurable by operating mode
Elevation coverage	Electronically controlled; installation- and mission-configurable

2. Primary model input set

Parameter	Representative value
Peak transmit power, Pt	750 kW
Pulse repetition frequency, PRF	300 Hz nominal
Transmit pulse width, tau	2.15 us
Antenna peak gain, G	37 dBi nominal at boresight
Azimuth beamwidth / search-cell resolution	2.8 deg nominal
Elevation beam spacing	2.23 deg nominal
Receiver noise figure	4.0 dB nominal
System loss allowance	6.0 dB representative end-to-end
Nominal range resolution	323 m

Surveillance Radar Modeling Notes

3. Reporting and signal-processing assumptions

Parameter	Representative value	Use in exercise
Range reporting resolution	323 m	Baseline range-reporting increment.
Azimuth reporting resolution	2.8 deg	Baseline azimuth-reporting increment.
Elevation reporting resolution	2.23 deg	Baseline elevation-reporting increment.
Range-rate reporting increment	100 m/s	Baseline range-rate-reporting increment.
False-alarm design point	1 x 10^-7 per resolution cell	Baseline design point; detection threshold remains configurable.
Unambiguous range (nominal)	Approx. 500 km / 270 nmi	Derived from the nominal 300 Hz PRF.
Pulse waveform	Coherent pulsed	Baseline waveform type.

4. Antenna and RF subsystem

Antenna assembly

Planar GaN transmit/receive AESA mounted on a rotating pedestal.

Peak gain	**37 dBi at boresight**
Azimuth resolution	**2.8 deg nominal**
Elevation spacing	**2.23 deg nominal**
Azimuth coverage	**360 deg continuous**

RF and waveform baseline

Carrier frequency	**2.9 GHz**
Peak power	**750 kW**
Pulse width	**2.15 us**
Nominal PRF	**300 Hz**
Noise figure	**4.0 dB**
System loss	**6.0 dB**

5. Scenario configuration notes

Item	Configuration status
Reporting range	Mission- and site-configurable; not fixed by the baseline hardware specification.
Mechanical scan rate	Configurable by operating mode.
Elevation limits / search volume	Configured for the installation and mission.
Detection threshold	Scenario- and processing-configuration dependent.
Target model / RCS	Selected separately for the analysis scenario.
Site and environmental losses	Selected separately; radome, processing, and installation losses may refine the 6 dB baseline.

Specifications are representative baseline values and subject to change without notice.